

HOTEL REVENUE DATA ANALYSIS AND VISUALIZATION USING PYTHON

**A SIX WEEKS INDUSTRIAL TRAINING REPORT IN THE PARTIAL
FULFILLMENT FOR THE AWARD OF THE DEGREE**

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IN

COMPUTER SCIENCE & ENGINEERING

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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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HOSHIARPUR – 146104, PUNJAB (INDIA) JUNE, 2026**

CERTIFICATE

This is to certify that the **Six Weeks Summer Training Report** entitled “**Python and Data science** ” has been successfully completed by **Ms. Harman Marwaha** University Roll No.**2449160**, a student of **Bachelor of Technology in Computer Science & Engineering**, Rayat Bahra Institute of Engineering & Nanotechnology, Hoshiarpur.

The project report has been prepared during the **Six Weeks Summer Training** conducted at **07Services** under my guidance and supervision. The work presented in this report is original and fulfills the requirements of the Summer Training as prescribed by the Institute.

To the best of my knowledge and belief, this project has not been submitted, either wholly or partially, to any other university or institution for the award of any degree or diploma.

This report is recommended for acceptance as a partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering** .

Training Guide**Head of Department**

Signature: _____

Signature: _____

Name: _____

Name: _____

Date: 25 - July - 2026**Place:** Hoshiarpur

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Finally, I sincerely thank my parents, family members, and friends for their unconditional support, motivation, and encouragement throughout the training period and during the preparation of this project report. Their constant inspiration helped me complete this work successfully.

I am grateful to everyone who directly or indirectly contributed to the successful completion of my Summer Training and this project report.

Date: 25- July - 2026

Place: Hoshiarpur

Student Signature: Harman Marwaha

DECLARATION

I hereby declare that the project report titled "**Hotel Revenue Data Analysis and Visualization using Python**" submitted to the Department of **Computer Science & Engineering, Rayat Bahra Institute of Engineering & Nanotechnology, Hoshiarpur**, is an original work carried out during my training at **07 Services**.

The work presented in this report has been completed by me under the guidance of my training mentor and faculty guide. The information, analysis, and implementation presented in this report are based on my learning, practical work, and project development during the training period.

I further declare that this report has not been submitted, either in full or in part, to any other university or institution for the award of any degree, diploma, or certificate. Wherever the work of others has been referred to,

appropriate acknowledgment has been made.

I understand that if any information provided in this declaration is found to be incorrect or misleading, I shall be solely responsible for the consequences.

Signed by:

Student Name: Harman Marwaha

Date: 24 – July - 2026

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CHAPTER 1: OVERVIEW OF TECHNOLOGY

This chapter provides a comprehensive overview of the technologies and libraries used in the Hotel Revenue Analysis project. Understanding these technologies is essential for implementing data science solutions effectively. The chapter covers Python programming language, data manipulation libraries (**Pandas, NumPy**), visualization libraries (**Matplotlib, Seaborn, Plotly**), and the **Streamlit** framework for building interactive web applications. Each technology is discussed in terms of its features, advantages, and specific applications within the context of our project.

1.1 Introduction to Data Science

Data science is an interdisciplinary field that combines **statistics, mathematics, computer science**, and domain expertise to extract meaningful insights from data. In today's digital age, organizations generate massive amounts of data daily, and the ability to analyze and interpret this data has become crucial for competitive advantage.

The hotel industry, in particular, generates vast quantities of operational data **including bookings, cancellations, guest reviews, occupancy rates, and revenue figures**. Traditional methods of analyzing this data using spreadsheets are time-consuming and limited in their analytical capabilities. Modern data science techniques enable hotel managers to uncover patterns, trends, and insights that drive business decisions.

Key Aspects of Data Science

- **Data Collection:** Gathering data from various sources including PMS systems, booking platforms, and guest feedback systems.
- **Data Cleaning:** Removing inconsistencies, handling missing values, and ensuring data .
- **Exploratory Data Analysis:** Understanding data structure, distributions, and relationships.
- **Statistical Analysis:** Applying statistical methods to test hypotheses and validate findings.
- **Data Visualization:** Creating visual representations to communicate insights effectively.
- **Machine Learning:** Building predictive models for forecasting and classification

Importance of Data Science

1. **Better Decision-Making** – Helps businesses make accurate decisions using data instead of guesswork.
2. **Predicts Future Trends** – Uses historical data to forecast future outcomes, such as sales and customer demands
3. **Improves Business Performance** – Identifies problems and opportunities to increase efficiency and profits.
4. **Enhances Customer Experience** – Analyzes customer behavior to provide personalized services and recommendations.
5. **Supports Healthcare** – Helps in disease prediction, medical research, and patient care.
6. **Detects Fraud** – Identifies suspicious activities in banking, finance, and online transactions.
7. **Automation** – Enables AI and machine learning to automate repetitive tasks.
8. **Used in Many Fields** – Applied in healthcare, finance, education, e-commerce, agriculture, transportation, and cybersecurity

Advantages of Data Science

- Better decision-making
- Improved business performance
- Accurate predictions
- Automation of tasks
- Better customer satisfaction
- Fraud detection
- Time-saving
- Cost reduction
- Improved efficiency
- Better resource utilization

1.2 Python Programming language

Python is a **high-level, interpreted programming language** known for its simplicity and readability. It has become the de facto standard for data science and machine learning applications. Python offers extensive libraries for data manipulation, numerical computing, visualization, and web development. The language's syntax is intuitive, making it accessible to both beginners and experienced developers. In the Hotel Revenue Analysis project, Python serves as the primary programming language for all data processing and analysis tasks. Python has emerged as the leading programming language for data science due to its simplicity, readability, and extensive ecosystem of libraries. Python's popularity in data science is attributed to several factors:

Advantages of Python:

- **Easy to Learn and Read:** Python's syntax is intuitive and resembles natural English, making it accessible to beginners.
- **Rich Ecosystem:** Python has a comprehensive collection of libraries specifically designed for data science.
- **Community Support:** A large and active community ensures continuous development and learning resources.
- **Flexibility:** Python supports multiple programming paradigms including procedural and object-oriented programming.
- **Performance:** Critical numerical operations can be optimized using compiled libraries like NumPy.
- **Integration:** Python easily integrates with other languages and tools, facilitating seamless data pipelines.

1.3 Data Analysis Libraries

1.3.1 NumPy Library

NumPy (**Numerical Python**) is the fundamental package for numerical computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. NumPy forms the foundation for many other data science libraries and is used in our project for numerical computations and array operations

Key Features

- N-dimensional array objects (ndarrays)
- Element-wise operations
- Linear algebra functions
- Random number generation
- Fourier transforms

Applications in Hotel Analysis

- Generating synthetic hotel data
- Performing statistical calculations
- Array manipulations for data preprocessing
- Computing aggregations and transformations

1.3.2 Pandas Library

Pandas is a powerful data manipulation and analysis library built on top of NumPy. It provides data structures **like DataFrames** that are ideal for working with tabular data. Pandas offers numerous functions for **data cleaning, transformation, aggregation, and reshaping**. Key features include **handling missing values, removing duplicates, filtering data, grouping operations, and merging datasets**. In our project, Pandas is extensively used for preprocessing the **hotel revenue dataset, handling missing values, and preparing data for analysis**.

Key Features

- DataFrame and Series data structures
- Data alignment and reshaping
- Handling missing data
- Grouping and aggregation
- Time series functionality
- Reading/writing data from various formats

Applications in Hotel Analysis

- Loading and exploring hotel revenue data
- Handling missing values and duplicates
- Filtering and sorting data
- Grouping by categories (month, guest type, region)
- Computing summary statistics
- Data transformation and feature engineering

1.4 Data Visualization Tools

1.4.1 Matplotlib Library

Matplotlib is the foundational plotting library in Python, providing a MATLAB-like interface for creating **static, animated, and interactive visualizations**. It provides a MATLAB-like interface for creating **plots, histograms, bar charts, scatter plots, and more**. Matplotlib offers fine-grained control over plot elements, making it suitable for publication-quality visualizations. The library integrates well with NumPy and Pandas, allowing seamless visualization of data. In the **Hotel Revenue Analysis project**, **Matplotlib is used for creating static visualizations of data distributions and trends**.

Visualization Types

- Line plots
- Scatter plots
- Bar charts
- Histograms
- Box plots
- Heatmaps
- 3D plots

Advantages

- Highly customizable
- Publication-quality output
- Extensive documentation
- Integration with other libraries

1.4.2 Seaborn Library

Seaborn is built on top of Matplotlib and provides a high-level interface for creating attractive statistical visualizations. Seaborn is a statistical data visualization library built on top of Matplotlib. It provides a **high-level interface for creating attractive and informative statistical graphics**. Seaborn simplifies the creation of complex visualizations with minimal code. It offers specialized plots for categorical data, **distributions, relationships, and time series**. The library integrates seamlessly with **Pandas DataFrames**. **In our project, Seaborn is used for creating statistical visualizations that reveal patterns and relationships in the data.**

Specialized Features

- Statistical estimation and plotting
- Categorical plots
- Distribution visualizations
- Correlation matrices

Advantages

- Beautiful default styles
- Simplified syntax for complex plots
- Integration with Pandas DataFrames
- Automatic color palettes

1.4.3 Plotly Library

Plotly creates interactive, web-based visualizations that allow users to hover, zoom, pan, and explore data dynamically. Plotly is a modern Python library for creating interactive, web-based visualizations. It supports a wide range of chart types **including scatter plots, line charts, bar charts, histograms, and more**. Plotly visualizations are interactive, allowing users to hover **for details, zoom, pan, and toggle data** series. The library is ideal for creating dashboards and web applications. Plotly Express provides a simplified interface for quick visualization creation. In the **Hotel Revenue Analysis project**, Plotly is used for creating interactive visualizations in the Streamlit dashboard

Interactive Features

- Hover tooltips
- Zoom and pan
- 3D visualizations
- Real-time updates

Visualization Types

- Scatter plots
- Line charts
- Bar charts
- Box plots
- Histograms
- Heatmaps
- Sunburst charts
- 3D scatter plots

Advantages

- Interactive exploration
- Web-ready output
- Professional appearance
- Mobile responsive

1.5 Streamlit

Streamlit is a modern web framework that enables data scientists to create interactive web applications without requiring HTML, CSS, or JavaScript expertise.

Key Features

- **Rapid Development:** Convert Python scripts into interactive web apps with minimal code changes.
- **Real-Time Updates:** Applications automatically refresh when code changes.
- **Interactive Widgets:** Sliders, buttons, dropdowns, and text inputs for user interaction.
- **Data Display:** Easy rendering of DataFrames, charts, and other visualizations.
- **Caching:** Performance optimization through intelligent caching.

Advantages for Hotel Analytics

- Non-technical users can explore data
- Real-time filtering and analysis
- Professional appearance without web development skills
- Easy to update and maintain
- Responsive design for mobile devices

CHAPTER 2: PROJECT REPORT WORK

2.1 Introduction

The **Hotel Revenue Analysis project** focuses on analyzing booking data from hotels to extract meaningful insights about revenue patterns, customer behavior, and operational efficiency. The hospitality industry generates vast amounts of data through **reservations, cancellations, guest interactions, and transactions**. This data, when properly analyzed, can reveal valuable patterns that help hotel managers make informed business decisions. The project utilizes advanced data science techniques to process, analyze, and visualize this data, providing actionable insights for revenue optimization and customer satisfaction improvement.

Python libraries like **Pandas, NumPy, Matplotlib, and Seaborn** are used to clean, process, analyze, and visualize the data. Different graphs and charts help identify revenue trends , cancellations , occupancy rate.

The project demonstrates how data science techniques can transform raw data into meaningful insights. It also improves analytical thinking and programming skills.

2.2 Problem Statement

The hotel industry operates in a **highly competitive and dynamic** environment. Effective revenue management is crucial for maximizing profitability and ensuring sustainable growth. However, many hotels struggle with accurately forecasting demand, identifying key revenue drivers, and understanding customer behavior due to the complexity and volume of operational data.

Specifically, challenges include:

- Lack of a centralized system for data analysis, leading to fragmented insights.
- Difficulty in identifying patterns and trends from raw, unstructured data.
- Inefficient manual processes for reporting and decision-making.
- Limited ability to perform predictive analysis for future demand and cancellations.
- Absence of interactive tools for exploring data and generating actionable insights.

Importance of the Project

- Understand hotel revenue and bookings
- Analyze historical hotel revenue data.
- Compare revenue performance across different months and guest types.
- Identify booking, cancellation, occupancy, and revenue trends.
- Visualize hotel data using graphs and charts.
- Learn practical applications of Python in Data Science.

2.3 PROJECT OBJECTIVE

The primary objective of this project is to develop a comprehensive data analysis and visualization solution for hotel revenue management. This solution aims to address the aforementioned challenges by providing a robust platform for data-driven decision-making.

The main objectives of this project are

- To analyze the hotel revenue analysis Dataset.
- To clean and preprocess the collected data.
- To remove missing values and duplicate records.
- To perform Exploratory Data Analysis (EDA).
- To understand hotel revenue and booking activities.
- To identify booking, cancellation , occupancy and revenue trends.
- To analyze bookings pattern
- To create attractive visualizations using Python.
- To understand hotel revenue problems
- To improve knowledge of data analysis and visualization techniques.

2.4 Methodology

The project follows a systematic process from data collection to visualization.

Step 1: Data Collection

The **Hotel Revenue** Dataset is collected in CSV format. The dataset contains information related to hotel revenue ,bookings, cancellations, guest types.

Step 2: Import Libraries

The required Python libraries are imported.

- Pandas
- NumPy
- Matplotlib
- Seaborn

These libraries help perform data analysis and visualization.

Step 3: Data Preprocessing

- **Data Loading:** Reading CSV files into Pandas DataFrames.
- **Missing Value Handling:** Identifying and imputing missing values using appropriate strategies (e.g., mean, median, mode, or forward/backward fill).
- **Check Duplicate :** Identifying and eliminating duplicate records to maintain data integrity.
- **Data Type Conversion:** Ensuring all columns have correct data types (e.g., converting date strings to datetime objects, numerical columns to appropriate numeric types).

Step 4: Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a crucial step in the data science workflow, allowing for a deep understanding of the dataset's characteristics, patterns, and potential issues. This phase involves summarizing the main characteristics of the data, often with visual methods.

Techniques Employed in EDA

- **Univariate Analysis:** Examining individual variables using histograms, box plots, and frequency distributions.
- **Bivariate Analysis:** Investigating relationships between two variables using scatter plots, grouped bar charts, and cross-tabulations.
- **Multivariate Analysis:** Exploring relationships among three or more variables, often with pair plots or 3D visualizations.
- **Correlation Analysis:** Quantifying the strength and direction of linear relationships between numerical variables using correlation matrices.

Step 5 : Visualization Implementation

Effective data visualization is key to communicating complex insights clearly and concisely. This project leverages a combination of Matplotlib, Seaborn, and Plotly to create a diverse range of visualizations.

Visualization Tools and Their Applications:

- **Matplotlib:** Used for generating static, publication-quality plots such as histograms, bar charts, and scatter plots.
- **Seaborn:** Employed for creating aesthetically pleasing statistical graphics, including heatmaps, violin plots, and pair plots.
- **Plotly:** Utilized for developing interactive, web-based visualizations that allow users to explore data dynamically through zooming, panning, and hovering.

Step 6: Interpretation

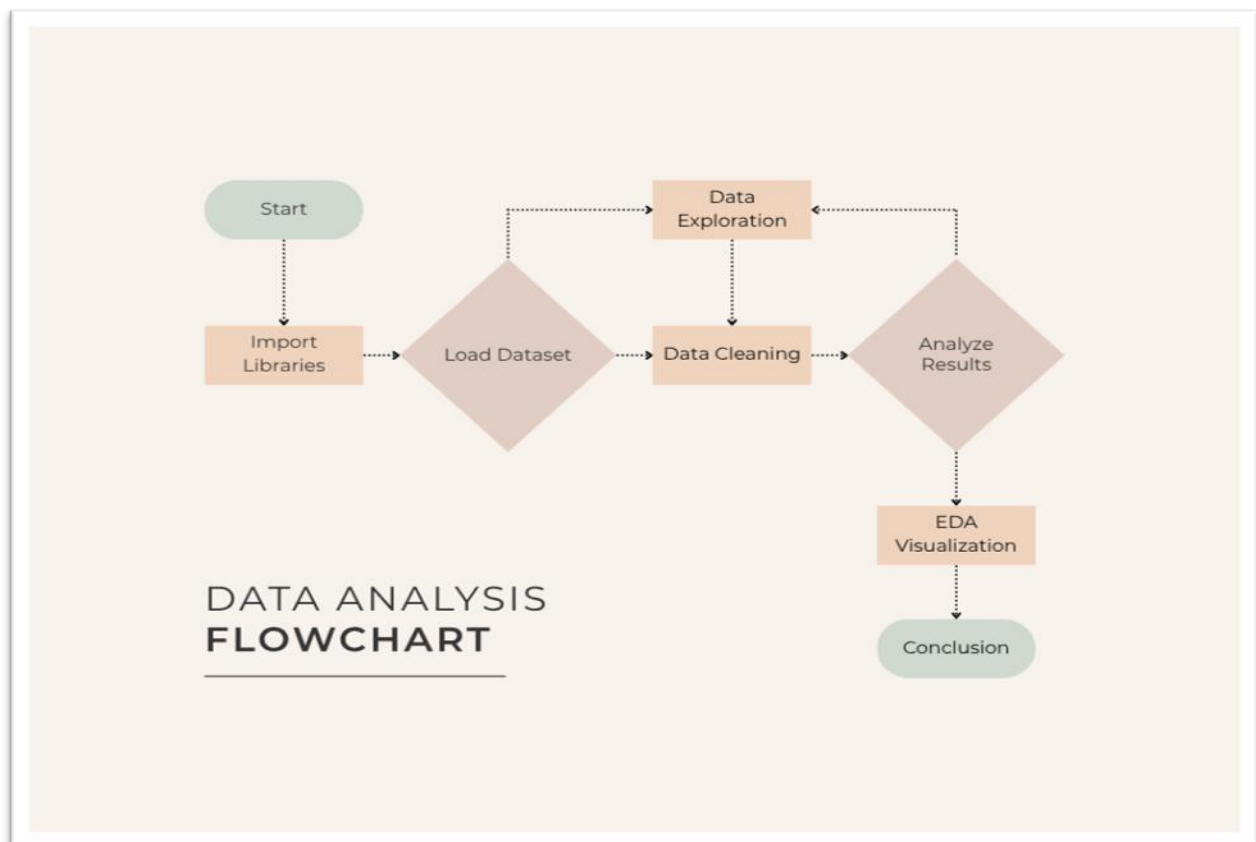
The graphs are interpreted to understand:

- Revenue growth trend
- Top performing guest types or market segments
- Monthly booking performance
- Revenue and occupancy distribution
- Cancellation and booking rates

2.4.1 Data Processing Flowchart

The data processing flowchart illustrates the sequential steps involved in transforming raw **hotel revenue** data into actionable insights. The process begins with data loading from CSV files, followed by **data cleaning and preprocessing steps including handling missing values, removing duplicates, and validating data types**. The cleaned data then undergoes exploratory data analysis to identify patterns and anomalies. Statistical analysis is performed to quantify relationships and trends.

Figure 2.1 : Data Processing Flowchart

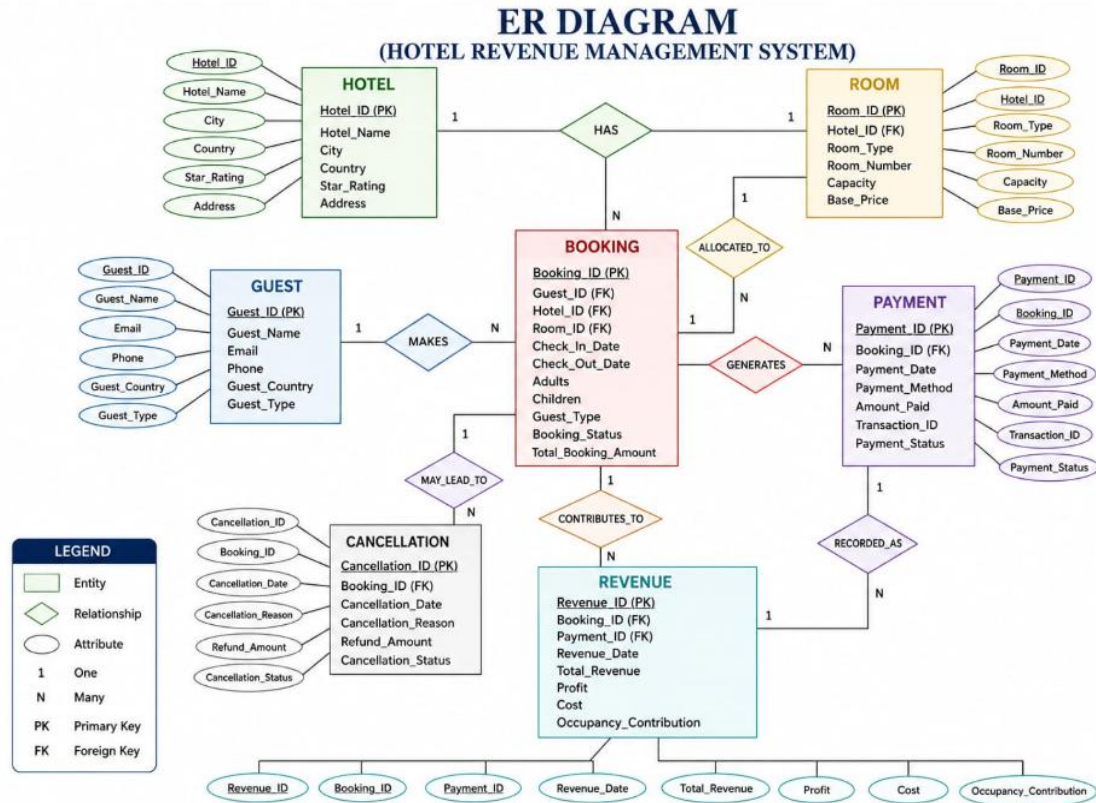


EXPLANATION OF FLOWCHART

The data analysis process starts by importing the required Python libraries and loading the dataset into the system. Next, the data is explored and cleaned by checking missing and duplicate values to ensure accuracy. Correlation analysis and Exploratory Data Analysis (EDA) visualizations are then performed

Hotel Revenue
to identify relationships, patterns, and trends in the data. Finally, the results are analyzed, meaningful conclusions are drawn, and the data analysis process is completed.

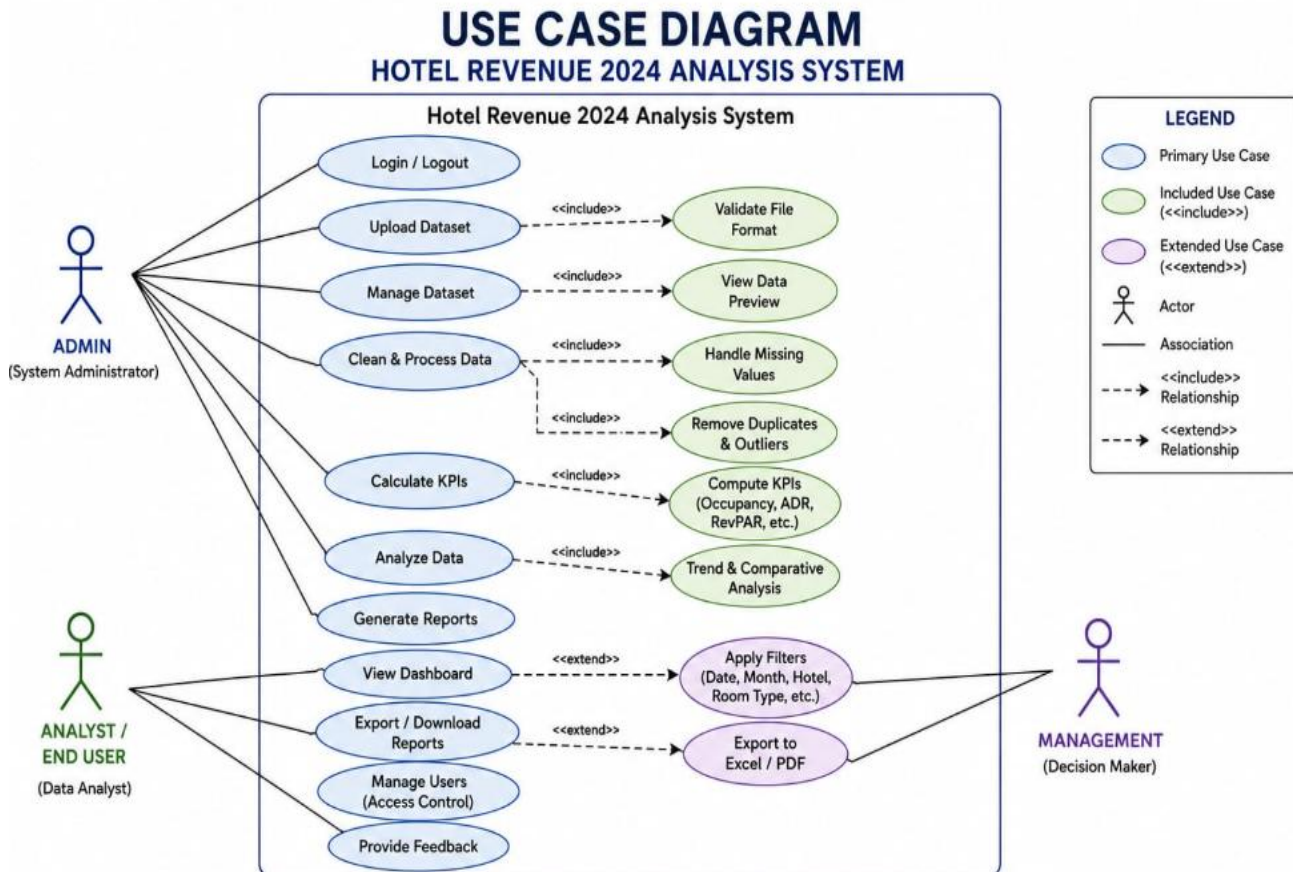
2.3.2 ER DIAGRAM



EXPLANATION OF ER Diagram

The **ER Diagram (Entity Relationship Diagram)** represents the database structure of the project. It contains the main entities: **Hotel, Room, Guest, Booking, Payment, Cancellation, and Revenue**, along with their attributes and relationships. A hotel contains multiple rooms, a guest can make multiple bookings, each booking is assigned to a room, bookings generate payment records, cancellations are linked to bookings, and every booking contributes to revenue. This diagram defines how data is organized and connected within the database.

2.3.3 USE CASE DIAGRAM

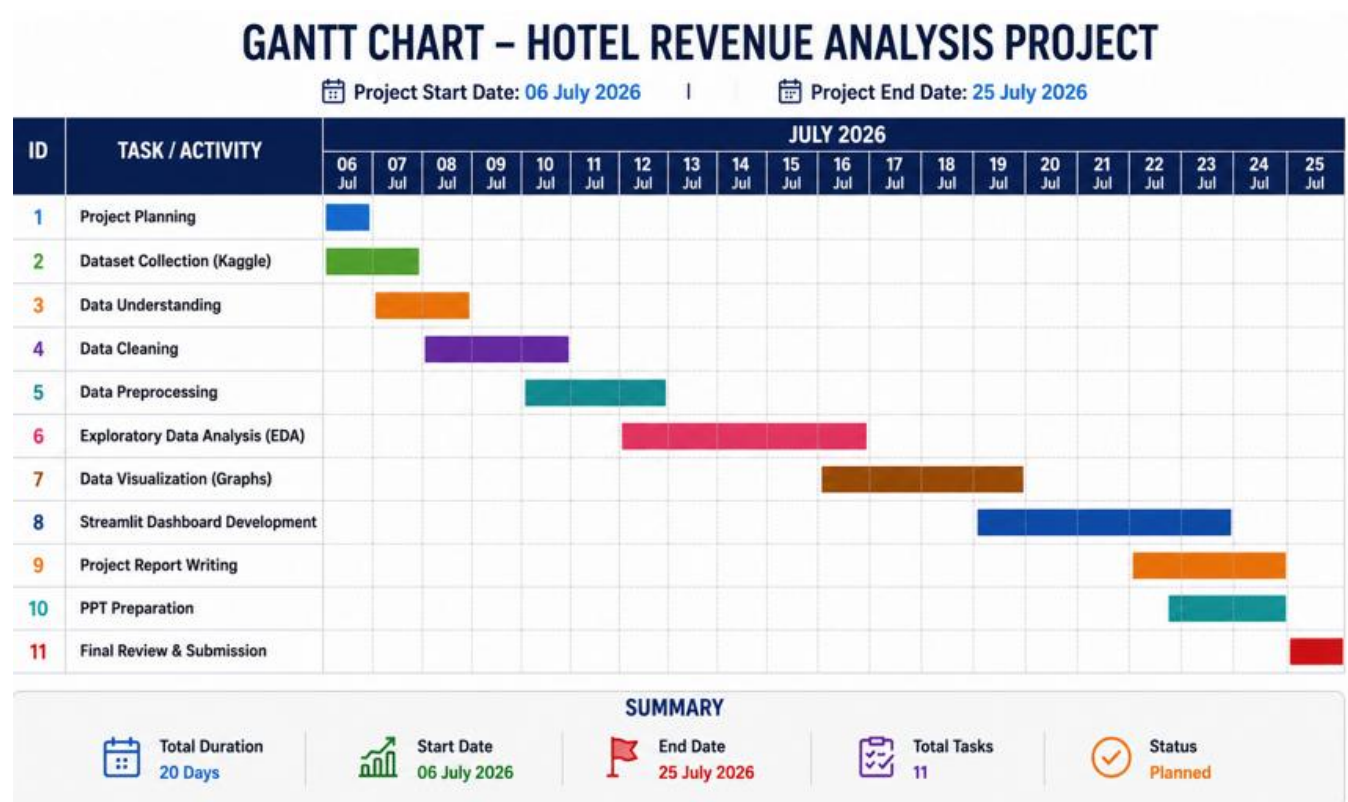


Explanation of Use case Diagram

The Use Case Diagram illustrates how users interact with the **Hotel Revenue 2024 Analysis System**. It includes three actors: Admin, Analyst/End User, and Management. The Admin manages the dataset, performs **data cleaning, calculates KPIs, analyzes data, and generates reports**. The Analyst views dashboards, exports reports, and provides feedback, while Management uses dashboards and reports to make informed business decisions. Overall, the diagram defines the system's main functions and user interactions.

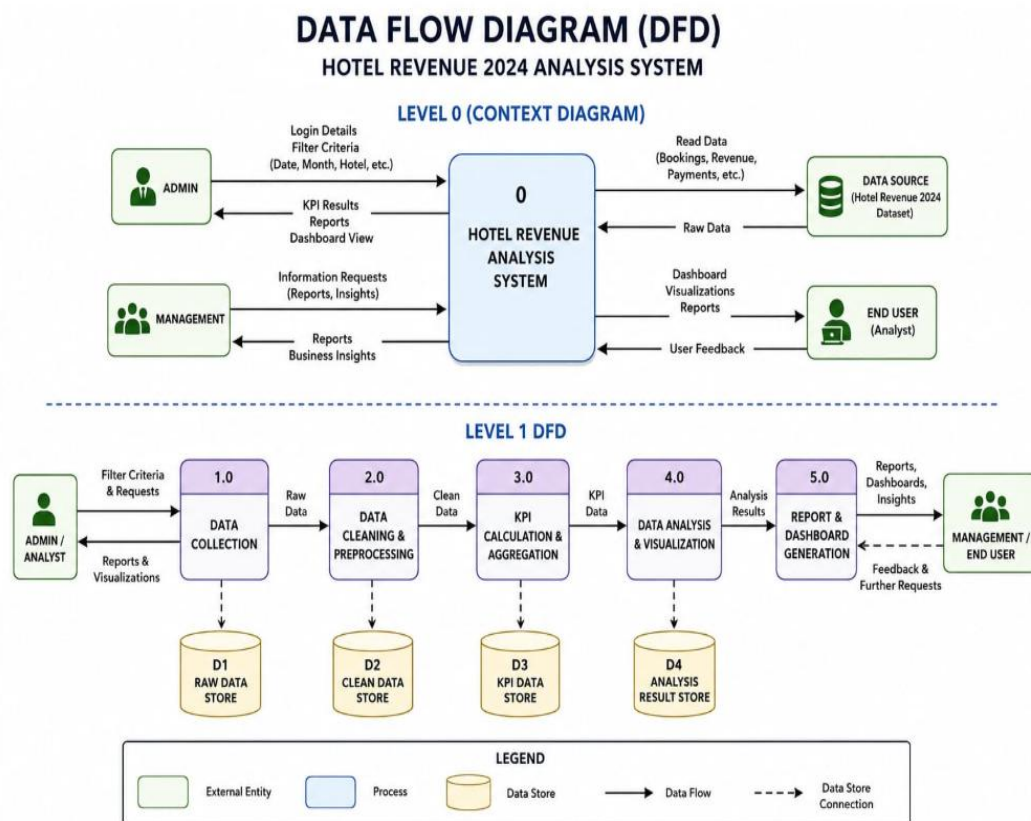
2.3.4 Project Gantt Chart

This Gantt chart represents the complete project timeline from 6 July to 25 July. We started with project planning on 6 July, followed by collecting the dataset from Kaggle and understanding the data. Next, we performed data cleaning to remove missing values and duplicate records, then carried out data preprocessing by converting data types and preparing the dataset for analysis. After that, we conducted Exploratory Data Analysis (EDA) to identify trends and patterns, followed by creating visualizations and graphs to present insights clearly. We then developed the Streamlit dashboard to display the analysis interactively. Finally, we prepared the project report, created the PowerPoint presentation, and completed the final review before submission on 25 July. This schedule helped us complete the project in a systematic and organized manner.



2.3.5 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) represents the flow of data through the system. External entities (users and admins) interact with the main analysis process. Data is read from raw data stores, processed, and results are stored in result databases. The DFD provides a clear visualization of how information moves through the system.



2.5 Hardware and Software Requirements:

Hardware Requirements

Processor: Intel Core i5

RAM: Minimum 4 GB (8 GB recommended)

Storage: At least 10 GB free disk space

Operating System: Windows 11,

Display: 1366×768 resolution or higher

Internet Connection: Required for installing libraries and updates

Software Requirements:

- Python 3.8 or higher
- Pandas 1.3.0 or higher
- NumPy 1.21.0 or higher
- Matplotlib 3.4.0 or higher
- Seaborn 0.11.0 or higher
- Plotly 5.0.0 or higher
- Streamlit 1.0.0 or higher
- Jupyter Notebook (optional, for development)

2.6 Scope of the Project:

This project focuses on analyzing historical hotel booking and revenue data to provide insights into various aspects of hotel operations. The scope includes data acquisition, cleaning, preprocessing, exploratory data analysis, statistical analysis, and the development of an interactive web dashboard. While the project aims to provide comprehensive insights, it has certain limitations.

Inclusions

- Data collection from a simulated hotel dataset.
- Extensive data cleaning and preprocessing using Pandas.
- Univariate, bivariate, and multivariate analysis.
- Visualization using Matplotlib, Seaborn, and Plotly.
- Statistical tests for hypothesis validation.
- Development of a Streamlit dashboard for interactive exploration.
- Generation of a detailed project report.

Exclusions

- Real-time data integration with live hotel systems.
- Predictive modeling (e.g., machine learning for demand forecasting or cancellation prediction).
- Deployment of the Streamlit application to a production server (focus on local deployment).
- Integration with external APIs for market intelligence.
- User authentication and authorization within the Streamlit app (focus on data presentation).

Chapter 3:USER GUIDE:

3.1 INTRODUCTION

The Hotel Revenue 2024 project is a data analysis application developed using Python in Jupyter Notebook. The main objective of this project is to analyze data related to Hotel Industry . The project processes the dataset, cleans unnecessary or missing values, performs statistical analysis, and creates graphical visualizations to help users understand trends in hotel revenue .

The project provides information about Booking ,Cancellations, Checkins , Total Revenue , Profit. Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the project transforms raw data into meaningful insights.

This chapter explains how users can install the required software, execute the project, understand each output screen, and interpret the results generated by the notebook.

3.1.1. Command Prompt

Description: The first step is to open the Command Prompt (CMD). Navigate to the folder where the project files are stored using the cd command. This ensures that the terminal is pointing to the correct project directory before launching Jupyter Notebooks

```
Microsoft Windows [Version 10.0.26100.3323]
(c) Microsoft Corporation. All rights reserved.

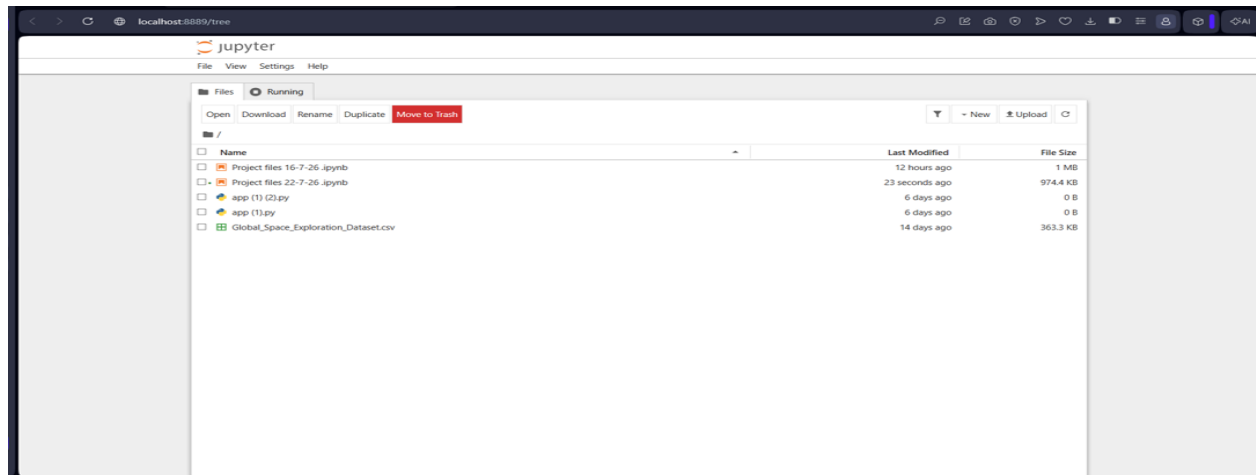
C:\Users\Komal Bhatia\OneDrive\Desktop\Data Science>Python -mnotebook
[1 2026-07-22 21:36:54.839 ServerApp] jupyter_lsp | extension was successfully linked.
[1 2026-07-22 21:36:54.849 ServerApp] jupyter_server_terminals | extension was successfully linked.
[1 2026-07-22 21:36:54.859 ServerApp] jupyterlab | extension was successfully linked.
[1 2026-07-22 21:36:54.869 ServerApp] notebook | extension was successfully linked.
[1 2026-07-22 21:36:55.477 ServerApp] notebook_shim | extension was successfully linked.
[1 2026-07-22 21:36:55.482 ServerApp] jupyter_lsp | extension was successfully loaded.
[1 2026-07-22 21:36:55.484 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[1 2026-07-22 21:36:55.498 LabApp] JupyterLab extension loaded from C:\Users\Komal Bhatia\AppData\Local\Programs\Python\Python314\Lib\site-packages\jupyterlab
[1 2026-07-22 21:36:55.498 LabApp] JupyterLab application directory is C:\Users\Komal Bhatia\AppData\Local\Programs\Python\Python314\share\jupyterlab
[1 2026-07-22 21:36:55.508 LabApp] Extension Manager is 'pypi'.
[1 2026-07-22 21:36:55.633 ServerApp] jupyterlab | extension was successfully loaded.
[1 2026-07-22 21:36:55.644 ServerApp] notebook | extension was successfully loaded.
[1 2026-07-22 21:36:55.646 ServerApp] Serving notebooks from local directory: C:\Users\Komal Bhatia\OneDrive\Desktop\Data Science
[1 2026-07-22 21:36:55.646 ServerApp] Jupyter Server 2.0.0 is running at:
[1 2026-07-22 21:36:55.647 ServerApp] http://localhost:8888/tree?token=688c3a251ef43c71f623a83c622bba9a0be08b2904241b98
[1 2026-07-22 21:36:55.647 ServerApp] http://127.0.0.1:8888/tree?token=688c3a251ef43c71f623a83c622bba9a0be08b2904241b98
[1 2026-07-22 21:36:55.647 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-07-22 21:36:55.703 ServerApp]

To access the server, open this file in a browser:
file:///C:/Users/Komal%20Bhatia/AppData/Roaming/jupyter/runtime/jpserver-20164-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=688c3a251ef43c71f623a83c622bba9a0be08b2904241b98
http://127.0.0.1:8888/tree?token=688c3a251ef43c71f623a83c622bba9a0be08b2904241b98
[1 2026-07-22 21:36:55.896 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-type
script-language-server, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language
-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageser
ver-bin, yamll-language-server
```

3.1.2. Launching Jupyter Notebook

Description: Run the command `jupyter notebook` in CMD. This starts the Jupyter Notebook server and automatically opens the notebook dashboard in the default web browser.

Purpose: To provide an interactive environment for executing Python code and visualizing outputs



3.1.3 Opening the Project Notebook

Description: From the Jupyter dashboard, open the notebook file Project files. The notebook contains all code cells required for data loading, preprocessing, visualization, and analysis.

Purpose: To access the complete project.

Hotel Revenue 2024 Data Analysis and Visualization using python

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Load dataset

```
[2]: df = pd.read_csv("hotelrevenue.csv")
df
```

	Date	Month	Weekday	Season	Holiday	Marketing_Spend	Revenue	Room_Revenue	Occupancy_Rate	ADR	...	Occupancy_Managed_Guests	RevPAR_All
0	01/01/2024	1	1	Winter	1	50000	80000	45000	0.75	120	...	75	4.5
1	02/01/2024	1	2	Winter	0	60000	85000	48000	0.78	125	...	78	4.6
2	03/01/2024	1	3	Winter	0	70000	90000	49000	0.80	130	...	80	4.7
3	04/01/2024	1	4	Winter	1	50000	80000	45000	0.75	120	...	75	4.5
4	05/01/2024	1	5	Winter	0	60000	85000	48000	0.78	125	...	78	4.6

3.1.4. Importing Required Libraries

Description: The first code cell imports the required Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Plotly. These libraries are essential for numerical computation, data manipulation, and visualization.

Purpose: To prepare the Python environment for data analysis

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

3.1.5. Loading the Dataset

Description: The dataset containing hotel revenue is loaded into a Pandas DataFrame using `pd.read_csv()`. This dataset includes hotel bookings, revenue, cancellations, occupancy, guest type.

Purpose: To make the project data available for analysis.

Load dataset

```
df = pd.read_csv("hotelrevenue.csv")
df
```

	Date	Month	Weekday	Season	Holiday	Marketing_Spend	Revenue	Room_Revenue	Occupancy_Rate	ADR	...	Occupancy_Managed_Guests	RevPAR_All
0	01/01/2024	1	1	Winter	1	50000	80000	45000	0.75	120	...	75	4.5
1	02/01/2024	1	2	Winter	0	60000	85000	48000	0.78	125	...	78	4.6
2	03/01/2024	1	3	Winter	0	70000	90000	49000	0.80	130	...	80	4.7
3	04/01/2024	1	4	Winter	1	50000	80000	45000	0.75	120	...	75	4.5
4	05/01/2024	1	5	Winter	0	60000	85000	48000	0.78	125	...	78	4.6
...	...	...	...	...	...	...	...	...	...	...	...	...	...
104	27/07/2024	7	5	Summer	0	50000	78000	43000	0.73	115	...	73	4.3
105	28/07/2024	7	6	Summer	0	55000	81000	44000	0.75	120	...	75	4.4
106	29/07/2024	7	7	Summer	0	65000	87000	46000	0.78	125	...	78	4.5
107	30/07/2024	7	1	Summer	0	70000	91000	48000	0.80	130	...	80	4.7
108	31/07/2024	7	2	Summer	0	50000	78000	43000	0.73	115	...	73	4.3

3.1.6.Dataset Information

Description: Commands such as `df.info()`, `df.shape`, and `df.describe()` display information about the dataset, including the number of rows and columns, data types, missing values, and statistical summaries.

Purpose: To understand the quality and structure of the data before analysis

Basic Information about the Dataset

```
1] : df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 109 entries, 0 to 108
Data columns (total 38 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                  109 non-null    str
1   Month                                109 non-null    int64
2   Weekday                              109 non-null    int64
3   Season                               109 non-null    str
4   Holiday                              109 non-null    int64
5   Marketing_Spend                      109 non-null    int64
6   Revenue                              109 non-null    int64
7   Room_Revenue                        109 non-null    int64
8   Occupancy_Rate                      109 non-null    float64
9   ADR                                  109 non-null    int64
10  RevPAR                               109 non-null    float64
11  Available_Rooms                      109 non-null    int64
12  Reserved_Rooms                      109 non-null    int64
13  Booking_Channel                      109 non-null    str
14  Guest_Type                           109 non-null    str
```

```
1] : df.shape
```

```
1] : (109, 38)
```

```
1] : df.columns
```

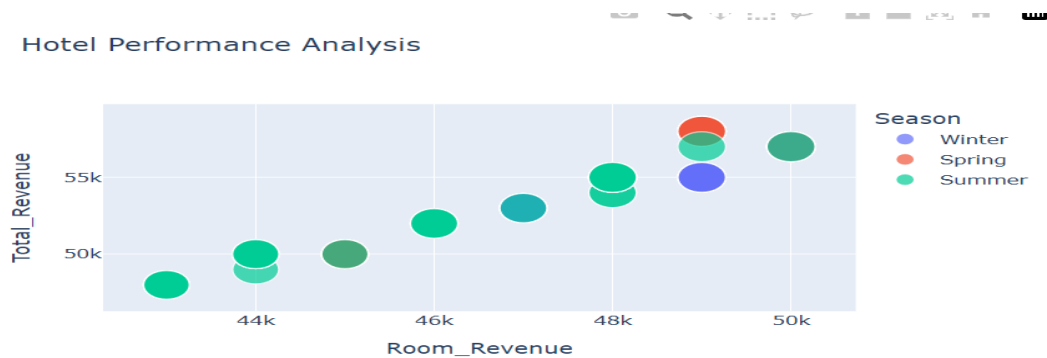
```
1] : Index(['Date', 'Month', 'Weekday', 'Season', 'Holiday', 'Marketing_Spend',
        'Revenue', 'Room_Revenue', 'Occupancy_Rate', 'ADR', 'RevPAR',
        'Available_Rooms', 'Reserved_Rooms', 'Booking_Channel', 'Guest_Type',
        'Market_Segment', 'Guest_Country', 'Complaints', 'Compliment',
        'Bookings', 'No_Shows', 'Cancellations', 'Checkouts', 'New_Bookings',
        'Checkins', 'Average_Review_Score', 'Revenue_Managed_Guests',
        'RevPAR_Managed_Guests', 'Occupancy_Managed_Guests', 'RevPAR_All',
        'Occupancy_All', 'Room_Revenue_All', 'Total_Revenue',
        'Operating_Expenses', 'Fixed_Costs', 'Variable_Costs', 'Total_Costs',
        'Profit'],
        dtype='str')
```

```
1] : df.dtypes
```

```
1] : Date                str
    Month              int64
    Weekday            int64
```

3.2 EDA ANALYSIS

3.2.1 Hotel Performance Analysis

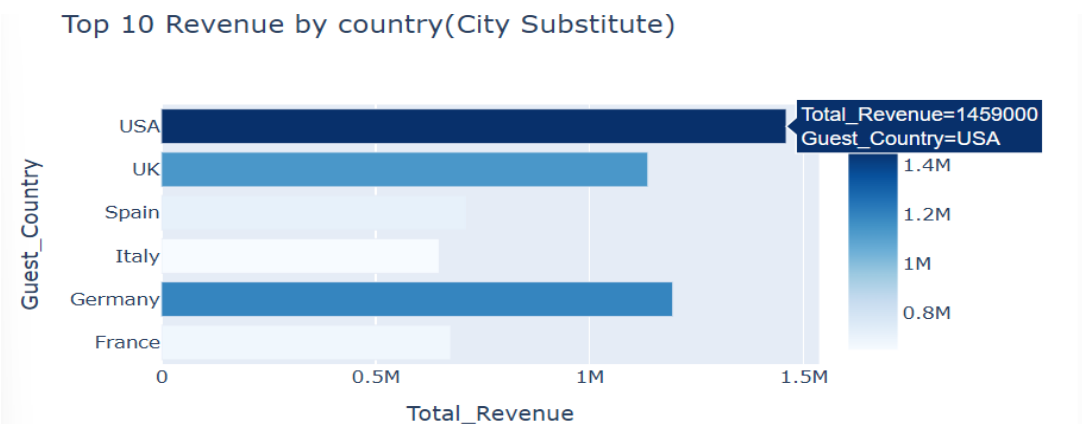


Hotel Performance Analysis

Compares Room Revenue with Total Revenue

colored by Season and sized by Occupancy Rate to analyze hotel performance.

3.2.2 Top Revenue by country(City Substitute)

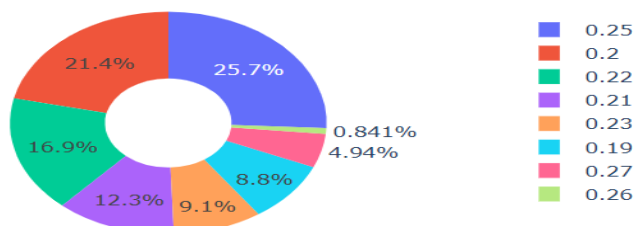


Key Insight

- Shows the total revenue generated by each guest country, helping compare country-wise revenue performance.
- Longer bars indicate higher revenue, while shorter bars indicate lower revenue contribution.

3.2.3 Revenue By Market segment

Revenue by Market Segment

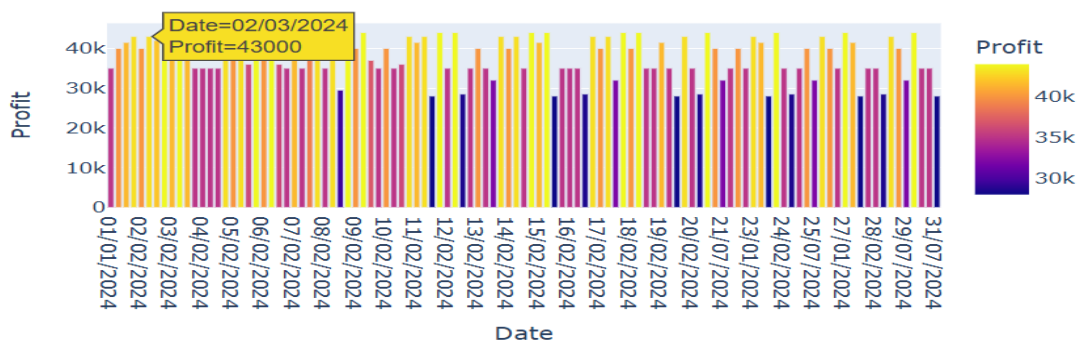


Key Insight

1. Shows the contribution of each market segment to the total revenue, making it easy to compare revenue shares.
2. Larger slices represent higher revenue contribution, while smaller slices indicate lower revenue contribution.

3.2.4 Overall Costs distribution

Overall Costs Distribution



KEY Insight

1. Shows the total profit earned on each date, helping identify daily profit performance.
2. Higher bars and yellow colors indicate higher profit, while shorter blue/purple bars indicate lower profit.

3.3 STREAMLIT CODE FOR FRONT END

3.3.1 Sample Code

```

C:\Users\HARMAN MARWAHA\OneDrive\Documents\GitHub\hotel-revenue dashboard > app.py
1 import streamlit as st
2 import pandas as pd
3 import numpy as np
4 import plotly.express as px
5 import plotly.graph_objects as go
6 import plotly.figure_factory as ff
7 from datetime import datetime
8 import base64
9
10 # =====
11 # 1. PAGE CONFIGURATION
12 # =====
13 st.set_page_config(
14     page_title="Hotel Revenue Analysis Dashboard",
15     page_icon="🏨",
16     layout="wide",
17     initial_sidebar_state="expanded"
18 )
19 # Initialize Session States for global access
20 if "feedback_data" not in st.session_state:
21     st.session_state.feedback_data = []
22
23 # =====
24 # 2. CUSTOM CSS
25 # =====
26 def apply_custom_css():
27     st.markdown("""
28     <style>
29     /* Main background and font */
30     .stApp {
31         background: linear-gradient(to right bottom, #f8f9fa, #e9ecef);
32         font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
33     }
34
35     /* KPI Cards */
36     .kpi-container {
37         display: flex;

```

3.3.2 Running Command

```

C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HARMAN MARWAHA\OneDrive\Documents\GitHub\hotel-revenue dashboard>python -m streamlit run app.py
2026-07-23 19:31:10.235 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.29.234:8501

Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.

C:\Users\HARMAN MARWAHA\OneDrive\Documents\GitHub\hotel-revenue dashboard\app.py:162: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
categorical_cols = df.select_dtypes(include=['object']).columns
2026-07-23 19:31:13.287 Please replace 'use_container_width' with 'width'.

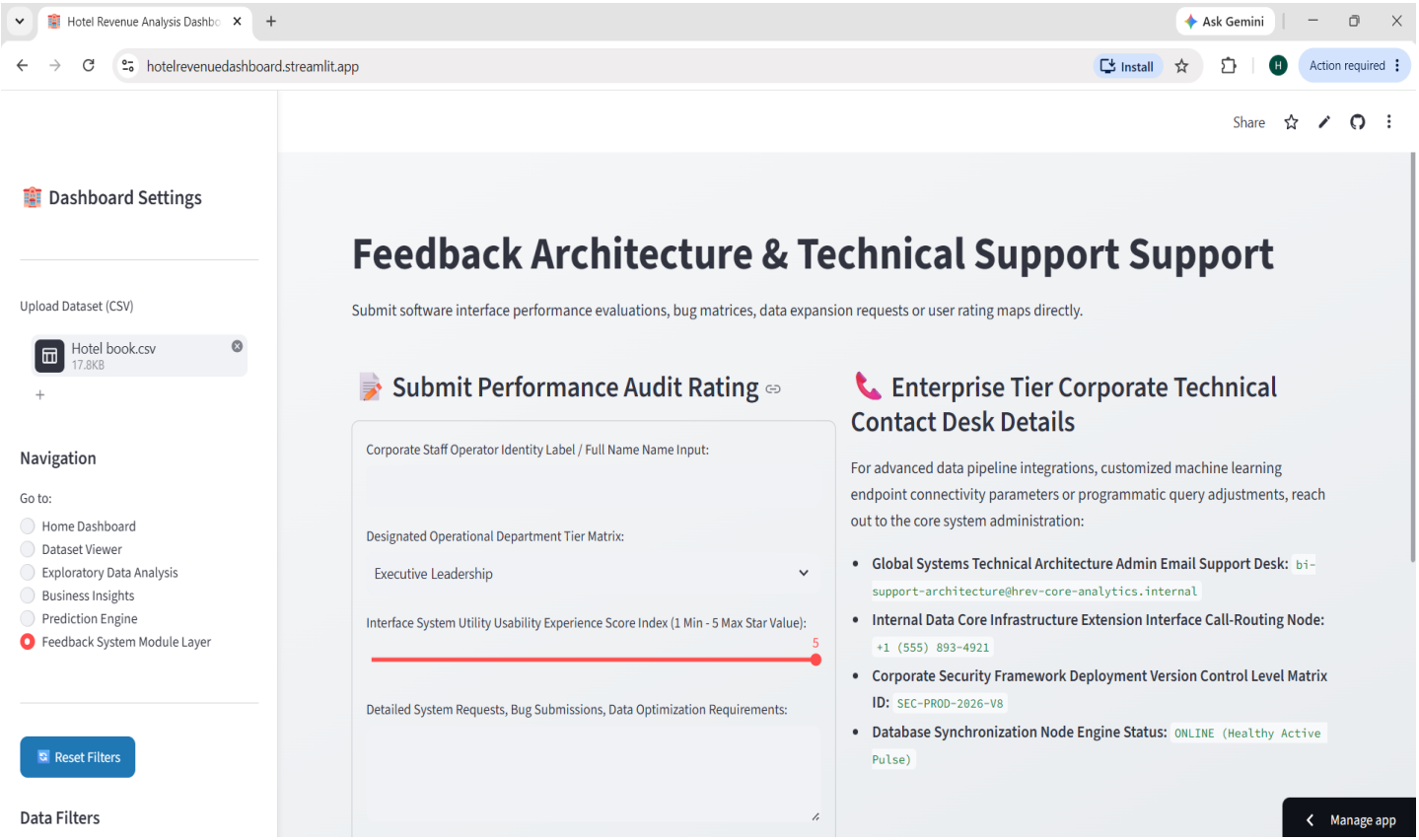
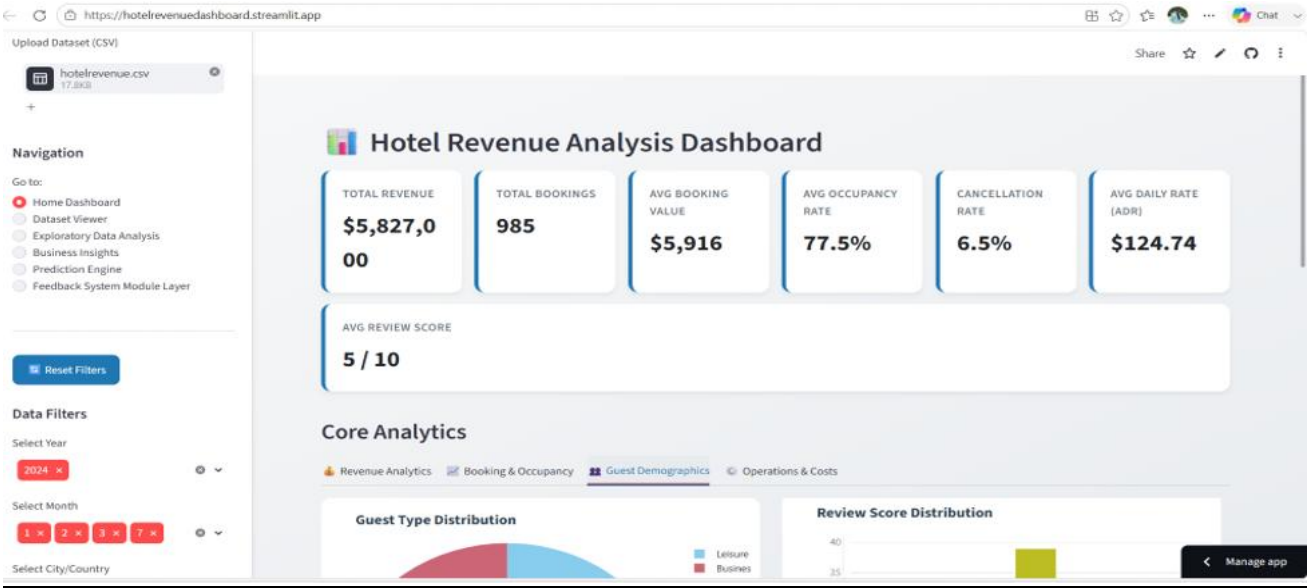
'use_container_width' will be removed after 2025-12-31.

For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.
2026-07-23 19:31:13.356 Please replace 'use_container_width' with 'width'.

'use_container_width' will be removed after 2025-12-31.

```

3.3.3 Front END <https://hotelrevenue dashboard.streamlit.app/>



3.4 Interpreting Results

The dashboard provides a wealth of information through its various visualizations and statistical summaries. To effectively interpret the results, users should focus on key metrics and trends.

Key Interpretations

Revenue Trends: Analyze line charts to identify seasonal patterns, growth, or decline in revenue over time.

Occupancy Rates: Monitor occupancy rates to understand demand fluctuations and operational efficiency.

Cancellations: Investigate cancellation rates by guest type, lead time, or market segment to identify areas for intervention.

Guest Behavior: Use categorical plots to understand preferences and booking patterns of different guest segments.

Profitability: Correlate revenue with costs and marketing spend to assess overall profitability and return on investment.

CHAPTER 4: CONCLUSION AND FUTURE WORK

4.1 Conclusion

This project successfully developed a comprehensive and interactive Hotel Revenue Analysis Dashboard using Python and its powerful data science libraries. By integrating NumPy, Pandas, Matplotlib, Seaborn, Plotly, and Streamlit, the solution provides a robust platform for data cleaning, preprocessing, exploratory data analysis, statistical analysis, and dynamic visualization of hotel operational data.

Key Achievements

- A streamlined data pipeline for efficient data preparation.
- In-depth insights into booking trends, revenue drivers, and customer behavior.
- Interactive visualizations that facilitate easy data exploration.
- A user-friendly web dashboard accessible to both technical and non-technical stakeholders.
- Actionable recommendations derived from statistical analysis to optimize revenue management strategies.

4.2 Future Scope

While the current project provides significant value, there are several avenues for future enhancements and extensions to further improve its capabilities.

Future enhancements to the Hotel Revenue Analysis project could include:

- Integration of predictive modeling to forecast future revenue and booking patterns
- Implementation of machine learning algorithms for customer segmentation and behavior prediction
- Development of real-time data integration to analyze current booking data as it arrives
- Enhancement of the dashboard with more advanced filtering and drill-down capabilities
- Integration with external data sources such as weather data, events, and competitor information

Potential Future Enhancements

- **Predictive Modeling:** Incorporate machine learning models (e.g., for demand forecasting, cancellation prediction, or dynamic pricing optimization) using libraries like Scikit-learn, XGBoost, or TensorFlow.
- **Real-time Data Integration:** Connect the dashboard to live hotel management systems or APIs for real-time data updates and monitoring.
- **Advanced Statistical Models:** Implement more sophisticated statistical models for deeper insights, such as time series forecasting with ARIMA or Prophet.
- **User Authentication and Authorization:** Add secure login mechanisms to the Streamlit dashboard to manage access for different user roles (e.g., administrators, revenue managers, marketing teams).
- **Customizable Reporting:** Allow users to generate custom reports with selected metrics and visualizations, exportable in various formats (PDF, Excel, PowerPoint).
- **Integration with External Data Sources:** Incorporate external data such as local events, weather forecasts, or competitor pricing to enrich the analysis.
- **Sentiment Analysis of Guest Reviews:** Utilize natural language processing (NLP) techniques to analyze guest reviews and extract sentiment, providing insights into customer satisfaction and areas for improvement.

A.3 Project File Summary

The project directory contains the following key files:

hotel_revenue_data.csv: Raw dataset
01_data_cleaning_preprocessing.py: Data cleaning script
02_exploratory_data_analysis.py: EDA script
03_statistical_analysis.py: Statistical analysis script
streamlit_hotel_analysis.py: Streamlit dashboard application
HOTEL_REVENUE_PROJECT_REPORT.docx:

report document

streamlit Dashboard:

<https://hotelrevenue dashboard.streamlit.app/>

BIBLIOGRAPHY

- McKinney, W. (2010). Data Structures for Statistical Computing in Python. In Proceedings of the 9th Python in Science Conference (Vol. 445, pp. 51-56).
- Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90-95.
- Waskom, M. L. (2021). Seaborn: statistical data visualization. Journal of Open Source Software, 6(60), 3021.
- Plotly Technologies Inc. (2015). Collaborative data science. Montréal, QC.
- Gabriel, A. (2020). Streamlit: The Fastest Way to Build Data Apps. O'Reilly Media.
- Abadi, M., et al. (2016). TensorFlow: Large-Scale Machine Learning on Heterogeneous Distributed Systems. arXiv preprint arXiv:1603.04467.
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794).
- Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In Advances in Neural Information Processing Systems (pp. 3146-3154).
- Virtanen, P., et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nature Methods, 17(3), 261-272.
- NumPy Developers. (2023). NumPy User Guide. <https://numpy.org/doc/stable/user/index.html>
- Pandas Development Team. (2023). Pandas User Guide. https://pandas.pydata.org/docs/user_guide/index.html
- Matplotlib Development Team. (2023). Matplotlib User Guide. <https://matplotlib.org/stable/users/index.html>
- Seaborn Development Team. (2023). Seaborn User Guide. <https://seaborn.pydata.org/tutorial.html>
- Plotly Development Team. (2023). Plotly Python Open Source Graphing Library. <https://plotly.com/python/>
- Streamlit Development Team. Streamlit Documentation. <https://hotelrevenue dashboard.streamlit.app/>

